

MATH70100 Bayesian Methods & Computation

Assessed Coursework 2

Due 9.00 a.m. on Tuesday 21 July 2026

Please submit a typed pdf report; **handwritten submissions will not be accepted**. For questions which require submission of computer code, present the relevant code segments inline as part of your report. Mathematical derivations and computer code should be kept concise, and derived expressions should be simplified whenever possible. Results from the lecture notes may be stated and used without proof.

This assignment must be attempted individually; your submission must be your own, unaided work. Candidates are prohibited from discussing assessed coursework, and must abide by [Imperial College's rules](#) regarding academic integrity and plagiarism. [Generative AI tools](#) (e.g., ChatGPT) may be used as a learning aid, but material generated by such tools must not be submitted as part of your assessed work, whether copied directly or adapted. Your submission must reflect your own understanding, reasoning, and presentation. Violations will be treated as an examination offence. Enabling other candidates to plagiarise your work constitutes an examination offence. To ensure quality assurance is maintained, departments may invite a random selection of students to an 'authenticity interview' on their submitted assessments.

This coursework addresses Bayesian model choice and model criticism, Gaussian process regression, non-conjugate generalised linear modelling, posterior computation using Stan, and Bayesian prediction and decision-making.

Question 1

Suppose real-valued responses $\mathbf{y} = (y_1, \dots, y_n)^\top$ are observed together with scalar covariates $x_1, \dots, x_n$. Consider the Gaussian linear model

$$\mathbf{y} \mid \beta, \sigma^2, X \sim \text{Normal}_n(X\beta, \sigma^2 I_n),$$

with conjugate prior

$$\beta \mid \sigma^2 \sim \text{Normal}_p(0, \sigma^2 \lambda^{-1} I_p), \quad \sigma^{-2} \sim \text{Gamma}(a, b),$$

where X is the design matrix and $\lambda, a, b > 0$ are known.

We consider the following three competing regression models:

$$\begin{aligned} M_1 : \quad \mathbb{E}[y_i \mid x_i] &= \beta_0 + \beta_1 x_i, \\ M_2 : \quad \mathbb{E}[y_i \mid x_i] &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2, \\ M_3 : \quad \mathbb{E}[y_i \mid x_i] &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3. \end{aligned}$$

The lecture notes show that under the conjugate Gaussian linear model,

$$\beta \mid \sigma^2, \mathbf{y}, X \sim \text{Normal}_p(m_n, \sigma^2 V_n),$$

where

$$V_n = (\lambda I_p + X^\top X)^{-1}, \quad m_n = V_n X^\top \mathbf{y},$$

and

$$\sigma^{-2} \mid \mathbf{y}, X \sim \text{Gamma}(a_n, b_n),$$

where

$$a_n = a + \frac{n}{2}, \quad b_n = b + \frac{1}{2} (\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top X m_n).$$

The corresponding marginal likelihood formula is also given in the lecture notes.

- a) For any two models M_i and M_j , write down the Bayes factor $B_{ij}(\mathbf{y})$ in terms of the model marginal likelihoods $p(\mathbf{y} \mid M_i)$, $p(\mathbf{y} \mid M_j)$ and explain how these quantities may be used to compare the three models M_1, M_2, M_3 . If equal prior probabilities are assigned to the three models, explain how to obtain posterior model probabilities. 3 Marks
- b) For posterior predictive checking under a chosen model with design matrix X , let $\mathbf{y}^{\text{rep}}$ denote a replicated dataset from the posterior predictive distribution, and define the discrepancy function

$$T(\mathbf{y}, \beta) = \sum_{i=1}^n (y_i - (X\beta)_i)^2.$$

Explain how a posterior predictive p -value based on T may be estimated by Monte Carlo, and what type of values would suggest possible lack of fit. 4 Marks

- c) The following data are observed (available on Blackboard as [xy.txt](#)):

i	1	2	3	4	5	6	7	8	9	10
x_i	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5
y_i	1.8	1.7	2.1	2.5	2.7	3.0	3.8	4.5	5.1	5.0

Assume $\lambda = 1$, $a = 2$, and $b = 2$. Write Python code to:

- i. compute the log marginal likelihood under each of M_1, M_2, M_3 ;
 - ii. compute the posterior probability of each model under equal prior model probabilities;
 - iii. identify the preferred model. 5 Marks
- d) Fit the preferred model from part c), and use posterior simulation to estimate the posterior predictive p -value based on the discrepancy statistic T . Briefly comment on whether the most probable model also appears to be an adequate model. 5 Marks
- e) Suppose that the posterior model probabilities place most mass on M_3 , but that the posterior predictive check from part d) suggests poor fit. Explain how this can happen, and briefly discuss two possible next steps a Bayesian analyst could take. 3 Marks

Question total: 20 Marks

Question 2

Suppose responses $\mathbf{y} = (y_1, \dots, y_n)^\top$ are observed at inputs $\mathbf{t} = (t_1, \dots, t_n) \in [0, T]^n$. Consider the Gaussian process regression model

$$y_i = f(t_i) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, \sigma^2),$$

where

$$f \sim \text{GP}(0, k),$$

with squared exponential kernel

$$k(s, t) = \alpha^2 \exp\left(-\frac{(s - t)^2}{2\rho^2}\right),$$

where $\alpha > 0$ and $\rho > 0$ are known.

- a) For a new input t^* , derive in matrix form the mean and variance of the posterior predictive distribution for $f(t^*) \mid \mathbf{t}, \mathbf{y}$. 5 Marks

- b) Briefly interpret the roles of the parameters α, ρ and σ^2 in the model. 3 Marks

- c) The following data are observed (available on Blackboard as [ty.txt](#)):

i	1	2	3	4	5	6	7	8	9	10	11	12	13
t_i	0.0	0.8	1.6	2.4	3.2	4.0	4.8	5.6	6.4	7.2	8.0	8.8	9.6
y_i	0.20	0.82	1.05	0.84	0.35	-0.18	-0.62	-0.48	0.02	0.74	1.10	0.88	0.41

Assume $T = 10$, $\alpha = 1$, $\rho = 1.5$, and $\sigma^2 = 0.05$. Write Python code to:

- plot the posterior mean regression function over $[0, T]$;
- construct pointwise 95% credible intervals for $f(t)$;
- estimate $\mathbb{P}(f(5) > 0 \mid \mathbf{t}, \mathbf{y})$.

5 Marks

- d) A decision-maker must choose a single number $d \in \mathbb{R}$ as a prediction of $f(5)$. Suppose the loss function is

$$\ell(d, f(5)) = \begin{cases} c_1(f(5) - d), & d \leq f(5), \\ c_2(d - f(5)), & d > f(5), \end{cases}$$

with $c_1, c_2 > 0$. Explain how the Bayes action depends on the posterior distribution of $f(5) \mid \mathbf{t}, \mathbf{y}$. Then, for $c_1 = 3$ and $c_2 = 1$, evaluate the Bayes action numerically for the observed dataset. 4 Marks

- e) Consider now two rival Gaussian process models for the same dataset:

- Model A uses the squared exponential kernel above with $\rho = 1.5$;
- Model B uses the same kernel but with $\rho = 0.4$.

Without carrying out a full numerical comparison, explain which model would typically be expected to give:

- a rougher posterior mean function;
- narrower uncertainty bands near the observations;
- a higher risk of overfitting the noise.

3 Marks

Question total: 20 Marks

Question 3

Consider binary responses $\mathbf{z} = (z_1, \dots, z_n) \in \{0, 1\}^n$ observed together with scalar covariates $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$. Suppose the data are modelled using a probit regression model:

$$z_i \mid \alpha, \beta \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p_i), \quad p_i = \Phi(\alpha + \beta x_i), \quad i = 1, \dots, n,$$

where Φ denotes the standard normal distribution function. Assume independent priors

$$\alpha \sim \text{Normal}(0, s_\alpha^2), \quad \beta \sim \text{Normal}(0, s_\beta^2),$$

where s_α^2 and s_β^2 are known.

a) Write down the posterior density $\pi(\alpha, \beta \mid \mathbf{x}, \mathbf{z})$ up to proportionality.

3 Marks

b) Explain why this model is non-conjugate, and why software such as Stan provides a convenient approach to posterior computation.

2 Marks

c) Write Stan code for this probit regression model. Your code should:

- declare the data and parameters;
- specify the prior and likelihood;
- include a generated quantities block returning the posterior predictive success probability q^* for a future observation at a supplied covariate value x^* ,

$$q^* = \Phi(\alpha + \beta x^*).$$

6 Marks

d) The following data are observed (available on Blackboard as [xz.txt](#)):

i	1	2	3	4	5	6	7	8	9	10	11	12
x_i	-2.5	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5	3.0
z_i	0	0	0	0	0	0	1	0	1	1	1	1

Take $s_\alpha^2 = s_\beta^2 = 25$ and $x^* = 0.75$.

Write Python code to fit the Stan model and obtain posterior samples.

3 Marks

e) Using your posterior sample, estimate:

- $\mathbb{E}[\beta \mid \mathbf{x}, \mathbf{z}]$;
- $\mathbb{P}(\beta > 0 \mid \mathbf{x}, \mathbf{z})$;
- $\mathbb{P}(z^* = 1 \mid x^* = 0.75, \mathbf{x}, \mathbf{z})$ for a future observation with covariate value $x^* = 0.75$.

3 Marks

f) Briefly compare probit and logistic regression as Bayesian models for binary data. Your answer should mention:

- one important similarity;
- one practical difference in interpretation or computation;
- one reason why the posterior predictive inferences for the two models may nevertheless be quite similar on many datasets.

3 Marks

Question total: 20 Marks

Total: 60 Marks